Trent W. Dawson, Ph.D.

Assistant Professor of Cyber Education and Director of the Governors Cyber Academy
College of Education & Human Performance
Dakota State University, 820 N Washington Ave, Madison, SD 57042
trent.dawson@dsu.edu | dtrentonvt@gmail.com
Academic Website | Google Scholar | ORCID

Education

2022	**Ph.D. in Curriculum Studies (Computer Science Education)** Louisiana State University, Baton Rouge, LA
2017	**M.Ed. in Instructional Design and Technology** Virginia Tech, Blacksburg, VA
2002	**B.B.A in Marketing** University of Oklahoma, Norman, OK

Research Interests

My interdisciplinary research explores educational pathways into computing and the preparation of educators across formal and informal learning environments. I design and investigate innovative educational programs and learning experiences that support all learners. This work includes research on computer science, artificial intelligence, and cybersecurity education to enhance computing self-efficacy, broaden participation in computing, and strengthen pathways into postsecondary education and the teaching profession.

Academic Appointments

Jun, 2026–Present	**Assistant Professor of Cyber Education and Director, Governors Cyber Academy**, College of Education and Human Performance, Dakota State University, Madison, SD
Jan, 2025–May, 2026	**Postdoctoral Fellow**, College of Education, University of Nevada, Las Vegas, Las Vegas, NV
Jun, 2024–Dec, 2024	**Postdoctoral Researcher and Adjunct Assistant Professor**, Department of Curriculum and Instruction, The University of Texas at Arlington, Arlington, TX

Professional Experience

Sep, 2023–May, 2024	**Educational Program Manager**, Department of Educational Programs, University of North Texas Health Science Center, Fort Worth, TX
May, 2023–Sep, 2023	**Assessment Coordinator**, Office of Assessment and Institutional Effectiveness, University of California, Merced, Merced, CA
2019–2021	**Graduate Assistant**, College of Education, Louisiana State University, Baton Rouge, LA

Refereed Journal Articles

[J3] **Dawson, T. W.**, Hoffman, A., Sahin, B., Fasching-Varner, K. J., & Hays, D. (in press). Program support and preservice teacher self-efficacy: A causal-comparative analysis of traditional and apprenticeship preparation pathways. *Action in Teacher Education*.

[J2] **Dawson, T. W.**, Butler, M., & Fasching-Varner, K. J. (2026). What counts as competent teaching? A systematic narrative synthesis of teacher competencies in K–12 education. *Teaching and Teacher Education*, *181*, 105684. https://doi.org/10.1016/j.tate.2026.105684

[J1] **Dawson, T. W.**, Hack, M., & Fasching-Varner, K. J. (2026). Onboarding nontraditional learners: An evaluation of an asynchronous online orientation in a teacher apprenticeship program. *The Journal of Continuing Higher Education*, 1–17. https://doi.org/10.1080/07377363.2026.2724232

Journal Manuscripts Under Revision

[R1] **Dawson, T. W.**, Joswick, C., Tobin, K., & Karadimou, O. (revise and resubmit). A pilot study of community college students' perceptions of computer science. *Discover Education*.

Journal Articles (in Preparation)

[Pr6] Schuh, M., **Dawson, T. W.**, Nash, J., Young, R., & De Jong, D. (in preparation). From apprentices to classroom teacher: Examining recent graduates' perceptions of preparedness and program effectiveness. [IRB-approved].

[Pr5] **Dawson, T. W.**, Berhans, A., & Cronin, K. (in preparation). Examining the relationship between high school summer cyber camp participation in and first-year matriculation at Dakota State University [IRB-approved].

[Pr4] Schneider, K., Schamber, B., Anderson, K., & **Dawson, T. W.** (in preparation). Examining teacher perceptions of summer professional development quality through five years of qualitative data. [IRB-approved].

[Pr3] **Dawson, T. W.**, Smith, K., & Welu, C. (in preparation). Exploring teachers' perceptions of cybersecurity and artificial intelligence following participation in cyber camp professional development [IRB-approved].

[Pr2] Schuh, M., Nash, J., **Dawson, T. W.**, Young, R., & De Jong, D. (in preparation). Exploring school principals' perspectives on graduates of a registered teacher apprenticeship program. [IRB-approved].

[Pr1] Salguero, M., **Dawson, T. W.** (in preparation). Systematic literature review examining institutional supports in shaping English learners' transition into first-year college.

Refereed Conference Proceedings

[C9] Smith, K., Welu, C., & **Dawson, T. W.** (accepted). From exposure to readiness: Teacher perspectives following a virtual summer cyber camp. In *Proceedings of the 58th Technical Symposium on Computer Science Education (SIGCSE)*, Sacramento, CA.

[C8] **Dawson, T. W.**, Morningstar, L., Bockman, R., & Sharma, R. (accepted). A factor analysis of community college students' attitudes about computer science and AI in introductory programming. In *Proceedings of the 57th Technical Symposium on Computer Science Education (SIGCSE)*, Sacramento, CA.

[C7] **Dawson, T. W.** (2026). Exploring self-doubt in community college computer science: Evidence from exploratory factor analysis. In *Proceedings of the 57th Technical Symposium on Computer Science Education (SIGCSE)*, pp. 1283–1284. St. Louis, MO. https://doi.org/10.1145/3770761.3777171

[C6] **Dawson, T. W.**, Sahin, B., Hack, M., & Fasching-Varner, K. J. (2026). Investigating pre-service teachers' perceived program support and its relationship to teaching self-efficacy across traditional and apprenticeship pathways. In *Proceedings of the Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA.

[C5] **Dawson, T. W.** (2026). Adapting computing attitudes surveys for community college students: Emerging constructs from factor analysis. In *Proceedings of the Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA.

[C4] Joshi, D., Joswick, C., Jocius, R., Albert, J., Blanton, M., Petrulin, B., & **Dawson, T. W.** (2025). Assessing elementary teachers' knowledge of integrated computational thinking. In *Proceedings of the 56th Annual Technical Symposium on Computer Science Education (SIGCSE TS 2025)*, pp. 1501–1502. Pittsburgh, PA. https://doi.org/10.1145/3641555.3705131

[C3] **Dawson, T. W.**, Calabrese, J. E., & Joswick, C. (2025). The learning styles zombie concept: A systematic review of NCTM publications. In *Proceedings of the Annual Meeting of the American Educational Research Association (AERA)*, Denver, CO.

[C2] **Dawson, T. W.** (2021). The block-based, text-based, and CS1 prepared. In *Proceedings of the 17th Annual Conference on International Computing Education Research (ICER)*, pp. 413–414. Virtual. https://doi.org/10.1145/3446871.3469777

[C1] Alegre, F., Moreno, J., **Dawson, T. W.**, Tanjong, E. E., & Kirshner, D. H. (2020). Computational thinking for STEM teacher leadership training at Louisiana State University. In *Proceedings of the Annual Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT)*, pp. 1–2. Portland, OR. https://doi.org/10.1109/RESPECT49803.2020.9272455

Refereed Presentations and Posters Sessions at Conferences

[P23] **Dawson, T. W.** (2026). *How to build your own computer science or cyber lesson.* Workshop presentation at the 2026 Annual DSU Cyber Camp for Teachers Conference, Dakota State University, Madison, SD.

[P22] **Dawson, T. W.**, Fluth, M., & Norberg, M. (2026). *Ready-to-implement cyber lessons for every grade level.* Presented at the South Dakota Society for Technology Annual Conference, Madison, SD.

[P21] **Dawson, T. W.**, Fasching-Varner, K. J., & De Jong, D. (2026). *Workforce-embedded pathways to teaching licensure: A tale of two statewide apprenticeship programs.* Paper presented at the 88th Annual Association for Continuing Higher Education (ACHE) Conference, Norfolk, VA.

[P20] **Dawson, T. W.** (2026). *A pilot study exploring attitudinal dimensions in community college computer science: Evidence from exploratory factor analysis.* Poster presented at the 57th ACM Technical Symposium on Computer Science Education (SIGCSE), St. Louis, MO.

[P19] **Dawson, T. W.**, Hoffman, A., Sahin, B., Hack, M., & Fasching-Varner, K. J. (2026). *Investigating pre-service teachers' perceived program support and its relationship to teaching self-efficacy across traditional and apprenticeship pathways.* Accepted presentation at the Annual Meeting of the American Educational Research Association (AERA), Los Angeles, CA.

[P18] **Dawson, T. W.** (2026). *Adapting computing attitudes surveys for community college students: Emerging constructs from factor analysis.* Accepted presentation at the Annual Meeting of the American Educational Research Association (AERA), Los Angeles, CA.

[P17] Joshi, D., Joswick, C., Jocius, R., Albert, J., Blanton, M., Petrulin, B., & **Dawson, T. W.** (2025). *Assessing elementary teachers' knowledge of integrated computational thinking.* Poster presented at the 56th ACM Technical Symposium on Computer Science Education (SIGCSE), Pittsburgh, PA.

[P16] **Dawson, T. W.**, Calabrese, J. E., & Joswick, C. (2025). *The learning styles zombie concept: A systematic review of NCTM publications.* Peer-reviewed paper accepted for presentation at the Annual Meeting of the American Educational Research Association (AERA), Denver, CO.

[P15] **Dawson, T. W.**, & Joswick, C. (2025). *Investigating the influence of K–12 computer science experience on community college students' interest in programming.* Peer-reviewed paper presented at the 2025 Annual Conference for the Southeast Educational Research Association (SERA), San Antonio, TX.

[P14] **Dawson, T. W.**, & Karadimou, O. (2025). *Navigating the Nevada Forward undergraduate apprenticeship program.* Poster presented at the Truckee Meadows Community College Annual EDU Night, Reno, NV.

[P13] Butler, M., Nolan, J., Karadimou, O., **Dawson, T. W.**, Hack, M., & Fasching-Varner, K. J. (2025). *The Nevada Forward Initiative: An apprenticeship program strengthening the teacher preparation pipeline.* Poster presented at the 2025 UNLV College of Education Annual Education Expo Conference, Las Vegas, NV.

[P12] **Dawson, T. W.**, & Manson, S. (2024). *Teaching strategies for academic medicine.* Session presented at the Texas Academy of Family Physicians Semiannual Conference, Austin, TX.

[P11] **Dawson, T. W.** (2024). *Demystifying nuances among variable types in statistics.* Session presented at the University of North Texas Health Science Center at Fort Worth's Graduate Certificate in Medical Education Annual Conference, Fort Worth, TX.

[P10] **Dawson, T. W.**, & Hollins, K. (2022). *Culturally responsive pedagogy to cultivate an inclusive learning environment.* Session presented at the Curriculum Camp Annual Conference at Louisiana State University, Baton Rouge, LA.

[P9] **Dawson, T. W.** (2022). *An evaluation of attitudes and beliefs toward computer science at community colleges.* Session presented at the Louisiana Educational Research Annual Conference (LERA), Lafayette, LA.

[P8] **Dawson, T. W.** (2021). *The block-based, text-based, and CS1 prepared.* Paper presentation at the 17th Annual ACM Conference on International Computing Education Research (ICER), August 16–19, 2021. Virtual event due to COVID-19.

[P7] **Dawson, T. W.**, Alegre, F., & Moreno, J. (2021). *A culmination of strategies for K–12 educators to assess students' CS knowledge.* Paper session presented at the UTeach STEM Educators Annual Conference. Virtual event due to COVID-19.

[P6] **Dawson, T. W.** (2021). *Comparing 15 computer science assessment strategies for pre-college learners: A*

systematic review of the CS literature. Paper session presented at the Curriculum Camp Annual Conference at Louisiana State University, Baton Rouge, LA.

[P5] **Dawson, T. W.**, Alegre, F., & Moreno, J. (2020). *A modified 5Es instructional model tailored for computer science educators*. Paper presentation accepted at the UTeach STEM Educators Annual Conference, Austin, TX. Conference canceled due to COVID-19.

[P4] Jenevein, S., Kirby, L., Harlow, L., Moreno, J., **Dawson, T. W.**, & Thompson, R. (2020). *Expanding computer science certification pathways*. Accepted for presentation at the UTeach STEM Educators Annual Conference, Austin, TX. Conference canceled due to COVID-19.

[P3] **Dawson, T. W.**, Alegre, F., & Moreno, J. (2020). *Adding a 6th "E" to the 5Es instructional model in computer science education*. Accepted for presentation at the American Association for the Advancement of Curriculum Studies (AAACS) Annual Conference, San Francisco, CA. Conference canceled due to COVID-19.

[P2] **Dawson, T. W.**, Fernando, A., & Moreno, J. (2020). *A modified 5Es instructional model tailored for computer science educators*. Session presented at the Louisiana Educational Research Annual Conference (LERA), Lafayette, LA.

[P1] **Dawson, T. W.**, & Reed, G. (2019). *Improving educational outcomes: Blending online with traditional face-to-face instruction*. Session presented at the Coalition of Adult Basic Education (COABE) Annual Conference, New Orleans, LA.

Research and Grant Experience

Co-Principal Investigator

Submitted **NSF E-CORE RII: BRIDGE: Building a Rural Innovation, Development, and Growth Ecosystem**
Dakota State University (subaward)
Proposed funding: $3,000,000 (overall proposal budget)

Principal Investigator — Grant Development

2026–Present **NSF CER: Building Supported Dual-Credit Pathways into Computing for Rural Youth**
Dakota State University
National Science Foundation, Computing Education Research (CER)
Role: Principal Investigator
Status: Proposal in preparation

Independent Research — Principal Investigator

Jul, 2025–Present **Community college computer science and cyber study**
University of Nevada, Las Vegas and College of Southern Nevada
Principal Investigator: Trent W. Dawson
UNLV and CSN IRB approved; second semester of data collection

Postdoctoral Research Experience

Jan, 2025–Present **Nevada Forward Initiative**
University of Nevada, Las Vegas
Principal Investigator: Kenny Varner
U.S. Department of Education, Augustus F. Hawkins Centers of Excellence Program
Awarded funding: $3,000,000

Jun, 2024–Dec, 2024 **Math + SEL + E**
The University of Texas at Arlington
Principal Investigator: Candace Joswick
U.S. Department of Education, SEED Grant
Awarded funding: $5,600,000 (Award #S423A220084)

Jun, 2024–Dec, 2024 **Unpacking Computational Thinking**
The Citadel and The University of Texas at Arlington
Principal Investigators: Jennifer Albert and Candace Joswick
National Science Foundation, DRK–12
Awarded funding: $2,550,527 (Award #2300322) and $443,801 (Award #2300323)

Grant Development

Oct, 2024 **Integrating K–12 AI/Cybersecurity into Computer Science**
The University of Texas at Arlington
Role: Grant Writer
Principal Investigator: Candace Joswick
National Science Foundation, DRK–12
Proposed funding: $820,000

Graduate Research Experience

May, 2020–May, 2022 **Baton Rouge Bringing Youth Technology Education Success**
Louisiana State University
Principal Investigator: Juana Moreno
National Science Foundation, CISE/CNS
Awarded funding: $5,000,000 (Award #1923573)

May, 2019–May, 2021 **Expanding and Strengthening the STEM Teacher Workforce**
Louisiana State University
Principal Investigator: Juana Moreno
U.S. Department of Education, SEED Grant
Awarded funding: $1,000,000 (Award #U423A180139)

Courses Taught and Other Teaching Experience

At Dakota State University

Spring, 2027 **CSC 804:** Computer and Cyber Science Research Methodology (Doctoral; scheduled)

At The University of Texas at Arlington

Fall, 2024 **EDUC 5394:** Understanding and Designing Classroom Research (Graduate)

At University of North Texas Health Science Center

Spring, 2024 **AMED 5600:** Scholarly Activity for Academic Health Professionals (Graduate)

At Louisiana Community Colleges

2018–2019 English and Mathematics (High School Equivalency)
South Louisiana Community College and Baton Rouge Community College

At Louisiana State University

2019–2022 Teaching Assistant
CSC 2700: Computer Science Teaching Methods (Undergraduate)
EDCI 7141: Studies in the Teaching of Mathematics in Secondary Schools (Graduate)
EDCI 2500: Knowing and Learning in Mathematics Education (Undergraduate)
EDCI 7821: Problems with Curriculum and Instruction (Graduate)

Professional Development Workshops

2025–Present Quarterly Pre-Service Teacher Professional Development, University of Nevada, Las Vegas

Fall, 2024 Monthly In-Service Teacher Professional Development, The University of Texas at Arlington

Supervision of Graduate Assistants

2026 **Maria Salguero**, PhD, Education Counseling, University of Nevada, Las Vegas

2025 **Sulabh Khadka**, PhD, STEM Education, University of Nevada, Las Vegas

2024 **Shaunak Amble**, MS, Computer Science, The University of Texas at Arlington

2024 **Keerthi Mandalapu**, MS, Data Science, The University of Texas at Arlington

Fellowships

2021 **International Computing Education Research (ICER) Doctoral Consortium Fellow**

Professional and Community Service

Service to Organizations

2024–2025 **Committee Member**, NSF BRICCS-TX, San Jacinto College
2023 **Committee Member**, DBER, University of California, Merced
2020–2022 **Executive Officer**, Curriculum Theory Project, Louisiana State University
2016–2017 **Treasurer**, Instructional Design and Technology Association, Virginia Tech

Service to Journals and Conferences

2026 **Manuscript Reviewer**, *Teaching and Teacher Education*
2026 **Session Chair**, Accessible Computing, ACM SIGCSE Technical Symposium
2025 **Manuscript Reviewer**, *Community College Review*
2025, 2026 **Proposal Reviewer**, American Educational Research Association (AERA), Division K, Section 6
2025, 2026 **Proposal Reviewer**, ACM SIGCSE Technical Symposium
2026 **Proposal Reviewer**, Society for Information Technology and Teacher Education
2024–2026 **Proposal Reviewer**, Southeastern Educational Research Association
2020–2022 **Proposal Reviewer**, Curriculum Theory Project Conference
2022 **Invited Speaker**, LSU Graduate Student Association Lunch and Learn
2021, 2022 **Session Chair**, Louisiana Educational Research Association Conference
2020, 2022 **Session Chair**, American Association for the Advancement of Curriculum Studies
2021, 2022 **Moderator**, American Association for the Advancement of Curriculum Studies

Service to Community

2024–2025 **Co-Chair**, Fundraising Committee, Dallas Roundup
2023–2024 **Chair**, Volunteer Committee, Dallas Roundup
2023–2024 **Chair**, Volunteer Committee, Black Men in White Coats Youth Summit

Technical Skills

R (advanced): Descriptive and inferential statistics, assumption testing, exploratory and confirmatory factor analysis (EFA/CFA), survey reliability analysis, missing data imputation, multiple regression with model fit evaluation, correlation analysis, and data visualization

ATLAS.ti (intermediate): Qualitative thematic coding

Qualtrics (expert): Survey design, distribution, and data export

Canvas (expert): Course design, asynchronous and hybrid instructional delivery

Scratch, Python, Java, Haskell (intermediate)

LaTeX (expert): Technical manuscript preparation and formatting

Academic Affiliations

Special Interest Group on Computer Science Education
American Educational Research Association
Southwest Educational Research Association
Computer Science Teachers Association
Association for Computing Machinery
International Computing Research Association
Computer Science for ALL